

Contact

+86 17315522050 (Mobile)
maoxq324@gmail.com

www.linkedin.com/in/maoxiangqian
(LinkedIn)

Top Skills

fluter
Web Engineering
Software Systems Engineering

Languages

English (Professional Working)
Chinese (Native or Bilingual)

Certifications

Software Designer (Intermediate Level)

xiangqian mao

Senior C++ & Qt Developer | 7+ Years Experience | Linux & Full-Stack Web | Remote Worldwide
Suzhou, Jiangsu, China

Summary

I am a senior Software Developer focused on C++ and Qt, with 7 years of experience developing embedded instruments and industrial machines (such as vision measurement systems and medical devices) for Linux and Windows environments, and 2 years of experience in full-stack web development. I have developed real-world projects including desktop applications using Qt and web applications with JavaScript, working across both frontend and backend. I am comfortable working in Linux and Windows environments and handling databases such as MySQL, PostgreSQL, and SQLite. My technical skills include C, C++, Qt, JavaScript, Python, Linux, and SQL. I enjoy building practical systems and continuously improving code quality and performance. I am currently seeking opportunities as a C++ Developer or Full-Stack Engineer, and I am open to remote roles worldwide or international positions.

Experience

Suzhou Vinno Technology

Software Engineer

March 2023 - March 2024 (1 year 1 month)

Suzhou

Designed and developed software systems for ultrasound endoscope devices, including system architecture, modular design and implementation, device integration, and database design (SQLite).

Designed and implemented core modules, including patient management, user management, medical record management, system configuration, and device management.

Compiled C++ libraries from source as needed and performed cross-compilation to produce ARM Linux and Windows binaries from an x86 Linux build environment, ensuring portability and reliability.

Led UX/UI design and development across Qt, Flutter, and web platforms, creating interactive and responsive user experiences.

Optimized image processing pipelines with GPU and multithreading, enabling fast and scalable handling of large-scale data.

Designed and implemented inter-process communication mechanisms for multiple processes to exchange data and coordinate tasks efficiently.

Conducted code reviews, created and maintained technical documentation, and developed unit tests to ensure maintainable, high-quality software.

Wuxi Research Institute of Huazhong University of Science and Technology (HUST)

Senior Software Engineer

October 2020 - March 2023 (2 years 6 months)

Wuxi

Designed and developed a 2D contour measurement tool for 3D point cloud processing (C++, Qt), including contour detection, extraction, and analysis.

Led system and architecture design, defining modular structure, data flow, component interactions, maintained technical document and unit tests.

Built a cross-platform (Windows/Linux) graphics system supporting interactive editing, geometric primitives, and real-time visualization.

Implemented fully interactive UI features, including movable graphics and measurement results with real-time updates.

Developed 2D contour measurement and feature extraction (length, radius, area, centroid) for industrial inspection.

Designed data persistence supporting multiple formats (JSON, TXT, XML, and databases).

Integrated with real-time 3D scanning systems and deployed in production environments.

Suzhou Huaxing Yuanchuang Technology Co., Ltd.

Software Engineer

July 2019 - October 2020 (1 year 4 months)

Suzhou

Designed and developed a suite of Qt-based tools for semiconductor testing, including a high-performance data management tool for handling and visualizing large-scale test data (millions of rows).

Processed and analyzed large-scale semiconductor test data, designing efficient methods to extract useful information from massive datasets.

Produced and maintained technical documentation, including system design documents, API specifications, and user manuals.

Education

Daqing Petroleum Institute

Bachelor's degree, Electronic Information Engineering · (September 2015 - June 2019)

东北石油大学

· (2015 - 2019)